SEO Review & 1-Week Execution Plan — 26 April 2026

Source data: Google Search Console export, last 90 days (2026-01-23 → 2026-04-22).
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Contents

1. Snapshot of where we are
2. Compressed 1-week execution plan
3. Book of work — user stories
4. Requirements consolidated (priority order)
5. Canonical-tag policy
6. Sitemap & robots policy
7. Reference — full multi-month strategy

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1. Snapshot of where we are

- 61 clicks / ~1,100 impressions / avg position **9.2** (page-1/page-2 boundary).
- 94% of impressions from India; brand + “guntur” geo modifiers dominate.
- Mobile = 75% of impressions, but only 4.66% CTR vs desktop 8.42% — mobile under-converts.
- Top page = `/` (879 imp, 5.23% CTR). `/services/` ranks well (pos 5.6) but converts at only 1.32% CTR.
- HTTP variant, trailing-slash variants, and `?page=…&lang=…` query variants are **all indexed separately** — link equity is being split.

High-intent demand we are leaking

Query	Impr	Pos	Clicks	Root cause
heart specialist in guntur	22	5.7	0	Title/meta CTR failure
echo and tmt test price	24	66	0	No pricing page exists
cardiologist in guntur	14	17.6	0	Stuck on page 2
best cardiologist in guntur	40	19.4	1	Stuck on page 2
2d echo test cost in guntur	8	12.3	0	No pricing content
eecp treatment in guntur	1	15	0	No `/services/eecp` page
kamalakar hospital reviews	3	11	0	No reviews page

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2. Compressed 1-week execution plan

The full multi-month strategy is preserved at the bottom of this document. The block below is what we commit to ship in the **week of 27 April – 3 May 2026**.

Day	Focus	Deliverable
Mon 27 Apr	Technical hygiene — canonicals, redirects, trailing slash	US-01, US-02, US-03
Tue 28 Apr	CTR rescue — titles & meta on top 4 pages	US-04, US-05
Wed 29 Apr	New page: diagnostics pricing	US-06
Thu 30 Apr	New page: EECP service + reviews block	US-07, US-08
Fri 1 May	FAQ schema + Physician/MedicalClinic JSON-LD upgrade	US-09, US-10
Sat 2 May	Sitemap + robots regeneration + GSC re-submit	US-11
Sun 3 May	Verification: Lighthouse, Rich Results, GSC URL inspection	US-12

Definition of done for the week:

1. Every indexed URL resolves to exactly one canonical (host, scheme, trailing-slash, query-string).
2. `/services/` and `/about/` SERP snippets rewritten — measured CTR delta over the following 14 days.
3. Two new high-intent pages live: pricing + EECP.
4. Sitemap regenerated, robots.txt reviewed, GSC re-pinged.

5. No regressions in Lighthouse (mobile ≥ 90 perf, ≥ 95 SEO).

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3. Book of work — user stories

Stories grouped by execution phase. **Phase 1 = this week.** Phase 2 and 3 are scheduled into May/June and listed here so the backlog is one document.

Priority key: P0 must ship this week, blocks other work or fixes a leak. P1 ships this week, high impact. P2 next 2–4 weeks. P3 longer-term / agentic-search work.

Phase 1 — Website-changeable, this week

US-01 — Single canonical host & scheme P0

> *As a search engine, I want one authoritative URL per page so I do not split ranking signals between `http://`, `https://`, `www`, and non-`www` versions.*

Acceptance criteria

- All `http://kamalakarheartcentre.com/*` requests 301 → `https://kamalakarheartcentre.com/*` at the CloudFront / origin layer.
- Every page emits `<link rel="canonical" href="https://kamalakarheartcentre.com{pathname}">` via `src/components/SEO.astro`.
- No page emits a canonical that points to a non-existent or redirected URL.
- Verified by `curl -I` on three sample pages.

Files touched: `cloudfront-functions/`, `src/components/SEO.astro`, every page that constructs `canonicalUrl` manually.

US-02 — One trailing-slash convention site-wide P0

> *As a crawler, I want `/services/ecg-echo` and `/services/ecg-echo/` to resolve to the same canonical URL so duplicate impressions stop appearing in Search Console.*

Acceptance criteria

- `astro.config.mjs` sets `trailingSlash: 'always'` (matches existing URLs already in GSC such as `/about/`, `/services/`, `/contact/`).
- A CloudFront function or origin rule 301s the no-slash variant to the slash variant for non-asset paths.
- All internal links (`<a href>`, sitemap entries, schema URLs, breadcrumbs) updated to the canonical form.
- `npm run build && grep -r 'href="/services/ecg-echo"' dist/` returns 0 hits.

Files touched: `astro.config.mjs`, `cloudfront-functions/`, `scripts/generate-sitemap.mjs`, internal link refs across `src/`.

US-03 — Strip indexable query-string variants of the homepage P0

> *As a search engine, I want `/?page=education`, `/?lang=te`, `/?page=education&lang=en`, `/?page=education&lang=te` to all canonicalise to a single primary URL so they stop being indexed as duplicates of `/`.*

Acceptance criteria

- Homepage emits `<link rel="canonical" href="https://kamalakarheartcentre.com/">` regardless of query string.
- The `?page=education` legacy pattern 301s to `/education/` (page already exists).
- The `?lang=te` legacy pattern 301s to `/te/`.
- Sitemap contains zero query-string URLs.
- Verified in GSC URL inspection for at least one previously-duplicated URL.

Files touched: `src/pages/index.astro`, `cloudfront-functions/`, `scripts/generate-sitemap.mjs`.

US-04 — Rewrite homepage `<title>` and meta description P1

> *As a Guntur patient searching for "best cardiologist in guntur", I want the SERP snippet to clearly identify Dr Kamalakar's specialism, location, and experience so I click through.*

Acceptance criteria

- Title ≤ 60 chars, leads with "Dr Kamalakar Kosaraju" + role + city.
- Description ≤ 155 chars, includes credentials (DM Cardiology), 30+ yrs experience, key services, Guntur.
- Meta refreshed in `src/content/site/en.yaml` and the Telugu equivalent.
- Sitemap `lastmod` updated (auto via existing build script).

Files touched: `src/content/site/en.yaml`, `src/content/site/te.yaml` (if exists), `src/pages/index.astro` if hardcoded.

US-05 — Rewrite title & meta on `/services/`, `/about/`, `/services/ecg-echo/`, `/services/hypertension-cholesterol/` P1

> *As a search-engine user, I want each landing page to promise a clearly differentiated answer in the SERP so I click on the right one.*

- `/services/` description names the actual procedures (Angioplasty, ECG/Echo, Pacemaker, EECP, BP/Cholesterol management).
- `/about/` title leads with credentials, links to author E-E-A-T story.
- `/services/ecg-echo/` includes the words "cost", "price", "Guntur" in the description.
- `/services/hypertension-cholesterol/` title + description recover its current 0% CTR at pos 5.

Files touched: `src/content/services/*.yaml`, `src/pages/services/[slug].astro`.

US-06 — New page: `/services/diagnostics-pricing/` P1

> *As a patient comparing test costs, I want a single page that lists ECG, 2D Echo, TMT, Holter, and angiogram pricing at Kamalakar Heart Centre so I can decide before visiting.*

- Live at `/services/diagnostics-pricing/`.
- Targets queries: `echo and tmt test price`, `2d echo test cost in guntur`, `echo tmt test price`, `angiogram cost in guntur`.
- Includes `MedicalProcedure` + `PriceSpecification` structured data per test.
- Linked from `/services/`, `/services/ecg-echo/`, homepage services grid.
- Internal nav: breadcrumb, prev/next.
- Sitemap regenerated (US-11).

Files touched: new `src/pages/services/diagnostics-pricing.astro` (or content yaml under `src/content/services/` if reusing `[slug].astro`), nav update, sitemap regen.

US-07 — New page: `/services/eecp/` P1

> *As a patient referred for EECP therapy in Guntur, I want a dedicated page describing the procedure, indications, and how to book at Kamalakar Heart Centre.*

- Live at `/services/eecp/`.
- Targets `eecp treatment in guntur` (currently pos 15 with no dedicated page).
- Uses the existing `[slug].astro` template via a new `eecp.yaml` content entry.
- Linked from `/services/` and the related-services block on `/services/heart-failure/`.

Files touched: new `src/content/services/eecp.yaml`, nav update.

US-08 — Reviews / testimonials block on homepage and `/about/` P1

> *As a prospective patient, I want to see real reviews so I trust the clinic before booking.*

- Reviews block visible on homepage and `/about/`.
- Markup uses `Review` items + `AggregateRating` on the `MedicalClinic` schema.
- Targets the `kamalakar hospital reviews` query.
- Source: at least 5 verifiable Google Business reviews quoted.

Files touched: `src/components/Reviews.astro` (new), `src/utils/schemas.ts`, page templates.

US-09 — FAQ schema on top service pages P1

> *As a search engine, I want structured FAQ Q&A on service pages so I can render rich results that lift CTR.*

- 4–6 FAQs on `/services/ecg-echo/`, `/services/hypertension-cholesterol/`, `/services/diagnostics-pricing/`, `/services/eecp/`.
- Marked up as `FAQPage` JSON-LD via `src/utils/schemas.ts`.
- Validates in Google Rich Results Test.
- Each answer ≤ 300 chars (LLM-extraction friendly).

Files touched: `src/utils/schemas.ts`, service yaml content, page templates.

US-10 — Upgrade site JSON-LD graph: `Physician` + `MedicalClinic` + `MedicalProcedure` P1

> *As an LLM grounding answer, I want one connected JSON-LD graph that ties Dr Kamalakar to the clinic and to each service so I can cite the site reliably.*

- Single JSON-LD graph using `@id` references: `Physician` → `worksFor` → `MedicalClinic` → `availableService` → `MedicalProcedure`.
- Author/byline on every blog post links to the `Physician` `@id`.
- `Physician` includes degrees, council registration, years of experience.
- Validates in Schema.org validator and Google Rich Results Test.

Files touched: `src/utils/schemas.ts`, layout templates, blog post template.

US-11 — Sitemap & robots.txt always rebuilt on content / route changes P0

> *As an operator, I want the sitemap and robots.txt to be regenerated automatically whenever a page is added, removed, renamed, or redirected so search engines never see a stale URL list.*

- `npm run build` invokes `scripts/generate-sitemap.mjs` (already exists) and the script output is part of the CI build artefact.
- The script reads pages and content collections — confirmed it picks up `diagnostics-pricing` and `eecp` automatically once US-06/07 ship.
- `public/robots.txt` reviewed: contains exactly one `Sitemap:` directive, points to the canonical HTTPS sitemap URL, and lists the AI crawlers.
- A short note added to `CLAUDE.md` and to the `deploy-verify` skill so this becomes a permanent pre-deploy gate: *"if you added/renamed/removed a page, rebuild sitemap and re-verify robots before deploying."*
- Post-deploy: GSC *Sitemaps* page shows the new sitemap successfully fetched.

Files touched: `package.json`, `scripts/generate-sitemap.mjs`, `public/robots.txt`, `CLAUDE.md`, `.claude/skills/deploy-verify/*` (if applicable).

US-12 — Verification & re-submission P0

> *As an SEO owner, I want post-deploy verification so I know the canonical, schema, and sitemap changes actually took effect.*

- Lighthouse mobile run on `/`, `/services/`, `/about/`, `/services/ecg-echo/`: SEO ≥ 95, Perf ≥ 90.
- Rich Results Test passes on home, services, blog, new pricing & EECP pages.
- GSC sitemap re-submitted; URL Inspection run for 3 previously-duplicated URLs to confirm canonical consolidation.
- Result captured in `SEO optimisation/26th April Review – verification.md`.

Phase 2 — Content gaps (May, weeks 2–4) P2

- **US-13** — Pillar page `/best-cardiologist-in-guntur/` (or homepage H1 stack rework) targeting the 40-impression query at pos 19.
- **US-14** — Author profile page `/team/dr-kamalakar/` with full E-E-A-T (degrees, registration #, fellowships, papers, hospital affiliations, photos).
- **US-15** — Telugu translation of top 4 service pages + about, linked from `/te/`.
- **US-16** — "Medically reviewed by" line + last-reviewed date on every blog post.

- **US-17** — Internal-linking pass across blog posts → pillar pages (3 topical clusters: interventional, diagnostics, chronic conditions).

Phase 3 — Agentic / LLM search (June onwards) P3

- **US-18** — Refresh `llms.txt` with pointers to highest-authority pages once new pages ship.
- **US-19** — Off-site NAP consistency: Practo, JustDial, Lybrate, Google Business — all match exactly.
- **US-20** — Wikidata entry for the clinic.
- **US-21** — Quarterly AI-mention audit across ChatGPT / Perplexity / Gemini / Google AI Overviews for the top 20 GSC queries. First run: end Q2 2026.
- **US-22** — Monthly GSC export diff vs this baseline so we can attribute lift to specific changes.

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4. Requirements consolidated — what changes on the website (priority order)

#	Requirement	Story	Priority	Repo area
1	All HTTP traffic 301s to HTTPS	US-01	P0	`cloudfront-functions/`
2	One trailing-slash convention enforced	US-02	P0	`astro.config.mjs`, CloudFront, sitemap
3	Homepage canonicalises away `?page=…` and `?lang=…` variants	US-03	P0	`src/pages/index.astro`, CloudFront
4	`<link rel="canonical">` present and correct on every page	US-01	P0	`src/components/SEO.astro`
5	Sitemap regenerated on every build; robots.txt reviewed	US-11	P0	`scripts/generate-sitemap.mjs`, `package.json`, `public/robots.txt`
6	Homepage title + meta rewritten	US-04	P1	`src/content/site/en.yaml`
7	Top 4 page titles + metas rewritten	US-05	P1	content yamls
8	New `/services/diagnostics-pricing/`	US-06	P1	`src/content/services/`, sitemap
9	New `/services/eecp/`	US-07	P1	`src/content/services/`, sitemap
10	Reviews block + `AggregateRating`	US-08	P1	`src/components/`, schemas
11	`FAQPage` schema on 4 service pages	US-09	P1	`src/utils/schemas.ts`
12	Connected JSON-LD graph (Physician ↔ Clinic ↔ Procedure)	US-10	P1	`src/utils/schemas.ts`
13	Verification: Lighthouse + Rich Results + GSC re-submit	US-12	P0	none (verification)

5. Canonical-tag policy

This rule applies from this week onward, enforced via `src/components/SEO.astro`:

1. Every page emits exactly one `<link rel="canonical">`.
2. The canonical URL is **absolute**, **HTTPS**, **bare domain** (no `www.`), and uses the chosen **trailing-slash convention** (`always`, per US-02).
3. The canonical never includes a query string unless the query string is the page's defining identifier (no current page qualifies).
4. Localised pages (`/te/...`) canonicalise to themselves and add `<link rel="alternate" hreflang="te">` and `<link rel="alternate" hreflang="en">` cross-references.
5. Paginated/filterable pages (none today) point canonical to the unfiltered base URL.
6. `og:url` matches the canonical exactly.
7. Build-time check: assert every `*.html` in `dist/` has a canonical tag whose URL resolves to itself.

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6. Sitemap & robots policy

1. `npm run build` always regenerates the sitemap. No manual sitemap edits.
2. The sitemap script is the single source of truth for indexable URLs — it reads from Astro pages + content collections, so any new `/services/*.yaml` or new `src/pages/*.astro` automatically appears.
3. After **any** of these, sitemap and robots are reviewed before deploy:
   - new page added
   - existing page renamed or path changed
   - page removed or 301-redirected
   - canonical convention changed
4. `public/robots.txt` is treated as code. Diff it on every PR. Exactly one `Sitemap:` line pointing to `https://kamalakarheartcentre.com/sitemap.xml`.
5. Post-deploy: hit `/sitemap.xml` and `/robots.txt` over HTTPS, confirm 200, then re-submit sitemap in GSC.
6. This rule is captured permanently in `CLAUDE.md` and the `deploy-verify` skill so future sessions cannot forget it.

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7. Reference — full multi-month strategy

Pillars for topical authority (Q2–Q3 2026):

1. Interventional cardiology (angioplasty / stents / angiogram) — pillar + 6 supporting articles.
2. Diagnostics & costs (ECG / Echo / TMT / Holter — with prices) — pillar + 4 articles.
3. Chronic conditions (hypertension / cholesterol / heart failure) — pillar + 6 articles.

Agentic-search readiness:

- E-E-A-T at the document level — author profile, medically-reviewed-by, last-reviewed dates.

- One connected `Physician` / `MedicalClinic` / `MedicalProcedure` JSON-LD graph (US-10).
- Short, factual TL;DR paragraphs (≤60 words) at the top of every clinical page — what LLMs lift verbatim.
- `FAQPage` blocks with ≤300-char answers.
- `llms.txt` updated as new authoritative pages ship.
- Off-site NAP consistency on Practo / JustDial / Lybrate / Google Business; encourage Google reviews.
- Wikidata entry for the clinic if/when notable.

Measurement cadence:

- Monthly: re-export GSC, diff against this baseline.
- Quarterly: AI-mention audit across ChatGPT / Perplexity / Gemini / Google AI Overviews for the top 20 GSC queries.

Document: `SEO optimisation/26th April Review.md` · Generated 26 April 2026.